

Vol 16 Chapter 7 — Bergman Kernel as Matrix-Coefficient Sum: Scaffolding v0.1

2026-06-05 Friday 13:15 EDT (date-verified)

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0. Chapter Thesis

The Bergman reproducing kernel $K_B(z, \bar{w})$ on $H^2(D_{IV}^5)$ decomposes as a sum of matrix coefficients across substrate K-types. This identification — Bergman kernel = K-type matrix coefficient sum — bridges Vol 16 Ch 4 (matrix coefficients = observables) and the substrate Hardy-space geometry developed in Chs 1+2.

Per Casey 12:30 EDT directive: substrate substantive content already exists across team's work; Vol 16 makes the linear-algebra-of-substrate explicit and portable.

1. Bergman Kernel Background

Standard form for D_{IV}^5

For the type IV Lie ball D_{IV}^n with rank 2 and complex dimension n (here $n = n_C = 5$), the Bergman kernel is:

$$\begin{aligned} K_B(z, \bar{w}) &= c_{FK} \cdot h(z, \bar{w})^{-(n+\text{rank})/2} \\ &= c_{FK} \cdot h(z, \bar{w})^{-7/2} \\ &= c_{FK} \cdot h(z, \bar{w})^{-g/2} \end{aligned}$$

where: $-h(z, \bar{w}) = 1 - 2\langle z, \bar{w} \rangle + (z \cdot z)(\bar{w} \cdot \bar{w})$ is the Jordan determinant $-(n_C + \text{rank})/2 = 7/2 = g/2$ substrate-natural exponent (Toy 3661 Helgason verification) - $c_{FK} \cdot \pi^{(9/2)} = 225$ (Lyra T2442 ratified)

Reproducing property

For any $f \in H^2(D_{IV}^5)$:

$$f(z) = \langle K_B(\cdot, \bar{z}), f \rangle_{FK}$$

Substrate substantive content

Per Lyra Sunday substrate operator framework: - K_B encodes substrate “now-state” projector onto holomorphic Hardy subspace - Bergman is the substrate-natural inner product on $H^2(D_{IV}^5)$ - $\kappa_{\text{Bergman}} = -n_C$ closed-form curvature scalar (Toy 3661)

2. K-Type Decomposition of $H^2(D_{IV}^5)$

Peter-Weyl style decomposition

$H^2(D_{IV}^5)$ decomposes under $K = SO(5) \times SO(2)$ action:

$$H^2(D_{IV}^5) = \bigoplus_{\{(\lambda_1, \lambda_2)\}} V_{-(\lambda_1, \lambda_2)} \otimes V_{-(\lambda_1, \lambda_2)}^*$$

where sum is over dominant K-weights.

Concretely (Lyra Chs 1-2 + Toy 3909 K-type catalog): - $V_-(0, 0)$ trivial (constants) - $V_-(1/2, 1/2)$ spinor (dim 4 = rank²) - $V_-(1, 0)$ vector (dim 5 = n_C) - $V_-(1, 1)$ adjoint (dim 10 = $2 \cdot n_C$) - $V_-(3/2, 1/2)$ muon spinor (dim 16 = $2^{N_C} \cdot \text{rank}$) - $V_-(2, 0)$ sym-trace (dim 14 = $2 \cdot g$) - $V_-(5/2, 1/2)$ tau spinor (dim 40 = $2^{N_C} \cdot n_C$) - ... and infinite tower of higher K-types

Reproducing kernel per K-type

Each isotypic component $V_-(\lambda_1, \lambda_2) \otimes V_-(\lambda_1, \lambda_2)^*$ has its own reproducing kernel $K^\lambda_B(z, \bar{w})$.

3. Matrix Coefficient Decomposition

Core formula

Per Peter-Weyl theorem applied to bounded symmetric domain Hardy space:

$$K_B(z, \bar{w}) = \sum_{\lambda} \{(\lambda_1, \lambda_2)\} (\dim V_-(\lambda_1, \lambda_2)) \cdot m^\lambda(z, \bar{w})$$

where $m^\lambda(z, \bar{w})$ is the matrix coefficient of K-type $V_-(\lambda_1, \lambda_2)$ at $(z, \bar{w})$.

Substrate Schur scalar

The Bergman kernel “diagonal” $K_B(z, \bar{z})$ on a K-orbit through z yields Schur scalar form:

$$K_B(z, \bar{z}) = \sum_{\lambda} (\dim V_{-\lambda}) \cdot s_{-\lambda}(K_B) \cdot \rho_{-\lambda}(z)$$

where $\rho_{-\lambda}(z)$ is the K-type density at z .

Substrate substantive content

Substrate Bergman is the sum of substrate K-type matrix coefficients weighted by K-type dimensions. Each K-type contributes its substrate Schur scalar to the substrate inner product structure.

4. Worked Example — Vacuum K-Type Contribution

Trivial K-type $V_-(0, 0)$:

$$\dim V_-(0, 0) = 1$$

$$m^{(0,0)}(z, \bar{w}) = 1 \quad (\text{constant function})$$

$$\text{Contribution: } 1 \cdot 1 = 1$$

This is the substrate vacuum contribution to the Bergman kernel. Substrate-natural unit normalization.

5. Worked Example — Spinor K-Type Contribution

$V_-(1/2, 1/2)$ spinor:

$$\dim V_-(1/2, 1/2) = 4 = \text{rank}^2$$

$$||V_-(1/2, 1/2)||^2 = 3\pi/2^g \quad (\text{Pochhammer, Toy 3695})$$

$$m^{(1/2, 1/2)}(z, \bar{w}) = \text{explicit polynomial in } z, \bar{w} \quad (\text{Faraut-Koranyi explicit form})$$

Spinor contribution to Bergman:

$$K_B^{\text{spinor}}(z, \bar{w}) = \text{rank}^2 \cdot m^{(1/2, 1/2)}(z, \bar{w})$$

Substrate substantive content: spinor cluster contributes 4-dim weight to substrate Bergman, scaled by spinor matrix coefficient.

6. Total Bergman as Generating Function

Hua-Schmidt formula

Per Hua + Faraut-Koranyi:

$$\begin{aligned} K_B(z, \bar{w}) &= c_{FK} \cdot h(z, \bar{w})^{(-g/2)} \\ &= c_{FK} \cdot \sum_{\{\lambda_1, \lambda_2\}} (\dim V_\lambda) \cdot \langle z^\lambda, \bar{w}^\lambda \rangle \end{aligned}$$

The series expansion of $h^{(-g/2)}$ in z and $\bar{w}$ gives the matrix-coefficient sum form. Each coefficient in the series is a substrate Schur scalar of corresponding K-type.

Substrate generating function reading

Bergman kernel = substrate generating function whose coefficients are substrate K-type matrix coefficients. Substrate content equals expansion of $h^{(-g/2)}$ in K-isotypic basis.

7. Heat Kernel = Mehler Family

Mehler kernel

For $\tau > 0$, define substrate Mehler kernel:

$$M_\tau(z, \bar{w}) = K_B(z, \bar{w})^\tau$$

K-type decomposition

By same K-type expansion:

$$M_\tau(z, \bar{w}) = \sum_\lambda (\dim V_\lambda) \cdot m_\tau^\lambda(z, \bar{w})$$

where m_τ^λ has substrate Schur scalar $\exp(-\tau \cdot C_2(\lambda)/\hbar_{BST})$.

Substrate substantive content

The Mehler family M_τ interpolates between identity ($\tau = 0$) and full Bergman kernel ($\tau = 1$). Each τ value gives substrate heat-evolved K-type matrix coefficients.

Per Toy 3905 substrate Mehler kernel framework: - Substrate diagonal matrix coefficient $\langle V_\lambda | M_\tau | V_\lambda \rangle = \exp(-\tau \cdot C_2(\lambda)/\hbar_{BST}) \cdot \|V_\lambda\|^2$ - Substrate off-diagonal vanishes for K-invariant Mehler

8. $\kappa_{Bergman} = -n_C$ Closed Form

Per Toy 3661 substrate Helgason application:

$$\kappa_{Bergman}(D_{IV}^5) = -n_C = -5$$

Substrate Bergman scalar curvature equals minus substrate spatial primary.

Matrix coefficient interpretation

Per Vol 16 Ch 4 Schur scalar framework, $\kappa_{Bergman}$ is the substrate Schur scalar of substrate curvature operator on Bergman kernel:

$$\kappa_{Bergman} = s(R_{Bergman}) = -n_C$$

Substrate substantive content

- $\kappa < 0$ substrate negatively curved (substrate-natural for Hermitian symmetric domain)
- $|\kappa| = n_C$ substrate primary
- Substrate curvature substrate-natural quantization in n_C units

Per Cal #34 STANDING Helgason closed form ratified (Toy 3661).

9. Cross-Anchor with Vol 16 Other Chapters

Ch 1 (Lyra) — Hilbert Space + Operator Algebra

Provides $H^2(D_{IV}^5)$ substrate Hardy space + substrate FK inner product. Ch 7 Bergman kernel = reproducing kernel for this Hilbert space.

Ch 2 (Lyra + Grace) — K-Type Spectral Decomposition

Provides K-type $\oplus$ decomposition of $H^2(D_{IV}^5)$. Ch 7 expresses Bergman as sum over K-types from Ch 2.

Ch 3 (Lyra) — Substrate Hall Algebra

Provides Hall algebra structure. Ch 7 Bergman matrix coefficients realize Hall algebra elements.

Ch 4 (Elie + Grace + Keeper) — Matrix Coefficients = Observables

Provides matrix coefficient framework. Ch 7 applies framework to Bergman kernel decomposition.

Ch 6 (Lyra) — Casey #14 Chirality Projection

Provides substrate K-type restriction $SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$. Ch 7 Bergman behavior under restriction.

Ch 8 (Keeper + Lyra) — Curvature Scalars in Operator Language

Uses Ch 7 $\kappa_{\text{Bergman}} = -n_C$ as central curvature observable.

10. Multi-Week K-Audit Gates

Per Cal #189 Brake 2 substrate-mechanism FORWARD multi-week gates:

1. Bergman matrix-coefficient sum series convergence rigorous (analytic question)
2. K-type matrix coefficients $m^\lambda(z, \bar{w})$ explicit Faraut-Koranyi forms per K-type
3. Mehler kernel M_τ K-type decomposition rigorous
4. $\kappa_{\text{Bergman}} = -n_C$ Helgason cross-anchor rigorous (Toy 3661 already substantive)
5. Cross-anchor with Lyra Vol 16 Chs 1+2 substantive content
6. Joint Lyra + Elie multi-week chapter development

11. Joint Lyra + Elie Coordination

Per Casey 12:30 EDT Ch 7 = joint Lyra + Elie:

- Lyra primary: substrate Hilbert space + K-type decomposition (Chs 1+2 anchors)
- Elie primary: substrate matrix coefficient + Pochhammer numerical (Ch 4 anchor)
- Joint: substrate Bergman series expansion explicit per K-type

This v0.1 establishes joint scaffolding. Lyra integration with substrate Hall algebra (Ch 3) + K-type spectral framework (Ch 2) pending. Multi-week joint chapter development.

12. Status

v0.1 SUBSTANTIVE Elie initial scaffolding for joint Ch 7. Vol 16 architectural sprint operational across Chs 1 (Lyra L21), 3 (Lyra F26 absorbing Hall paper), 4 (Elie + Grace + Keeper v0.2), 7 (this doc), 8 (Keeper outline planned), 9 (joint).

Per Casey 12:30 EDT + Keeper 13:00 EDT continuous work directive.

Multi-week residuals pending Lyra integration of Hardy-space + K-type decomp foundations.

— Elie, Friday 2026-06-05 13:15 EDT (date-verified)